Ayesh Sharuka

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PROFILE

AI Engineer with hands-on experience building and shipping production-grade AI systems end to end. Proven expertise in Python, FastAPI, and AWS (Bedrock, EC2, S3, Lambda, SageMaker), with delivered agentic systems, RAG pipelines, LLM-powered applications, and containerised cloud-native deployments. Experienced with LLM APIs (Claude, LLaMA, Gemini), agent orchestration frameworks (LangChain, LangGraph, MCP), and enterprise security patterns (JWT/ OAuth, RBAC) as production defaults. MSc in Artificial Intelligence (Dublin Business School), underpinned by a rigorous statistical foundation (BSc, University of Colombo). Adept at translating ambiguous business problems into scalable, maintainable AI products across the full system lifecycle.

WORK EXPERIENCE

EMUQ TECH Inc, *AI/ML Developer* 07/2024 – 09/2025 | United States

- Engineered a production-grade FastAPI service integrated with AWS EC2 and S3 to automate the end-to-end audio mastering pipeline, reducing manual processing time significantly.
- Built a dynamic SEO-optimised content generation platform using FastAPI and AWS Bedrock (Claude & LLAMA LLMs), generating keywords, meta descriptions, titles, and CTAs from structured business inputs.
- Designed Pydantic-validated APIs with JWT authentication and CORS controls, ensuring secure, extensible, and well-documented endpoints.

Dialog Axiata PLC, 02/2024 – 10/2024 | Sri Lanka

Industrial Training in Group Analytics & AI Platform

- Developed and deployed a Natural Language to SQL system using Google Gemini LLM, enabling non-technical users to query multi-table relational databases via natural language with measurable accuracy gains.
- Built an AWS CloudWatch resource-utilisation dashboard monitoring SageMaker pipelines; automated data ingestion via Lambda functions, saving pipeline data to S3 as CSV for downstream analytics.
- Created Python scripts to extract pipeline metrics (CPU, memory, disk, instance types) and generate custom CloudWatch widgets, contributing to cost reduction of automated ML pipelines.

Nildiya Hotel Restaurant & Bar, *Data Analyst Internship* 05/2023 – 02/2024 | Sri Lanka

- Collected, cleaned, and analysed operational and customer data using SQL and Python; findings directly informed service quality improvements and management reporting.
- Performed exploratory data analysis (EDA) on customer feedback datasets, identifying actionable trends that influenced strategic planning decisions.
- Produced visualisations and written reports presented to management to support data-driven decision-making.

EDUCATION

Dublin Business School, *MSc, Artificial Intelligence* 09/2025 – Present | Ireland

University of Colombo, 01/2020 – 05/2023 | Sri Lanka

B.Sc., Industrial Statistics & Mathematical Finance

SKILLS

Languages (Python, Java, SQL, TypeScript) | **AI / ML** (LLMs, Agentic Systems, RAG Pipelines, NLP, Computer Vision, Deep Learning, Generative AI, Prompt Engineering, Model Evaluation, Recommender Systems, Graph Neural Networks) | **Agent & Orchestration** (LangChain, LangGraph, Tool Calling, MCP, Sub-agents, Human-in-the-Loop) | **Frameworks & Libraries** (FastAPI, Spring Boot, Pydantic, TensorFlow/Keras, Pandas, NumPy) | **Cloud & Infrastructure** (Azure, AWS (Bedrock, EC2, S3, Lambda, SageMaker, CloudWatch), Terraform, Docker, Kubernetes, CI/CD, Microservices) | **LLM / AI APIs** (Claude API, LLaMA, Google Gemini API, AWS Bedrock, Azure OpenAI, OpenAI API) | **Security & Auth** (JWT, OAuth, RBAC, CORS, GDPR-aware Design) | **Engineering Practices** (REST APIs, Observability, | Logging & Tracing, Latency Optimisation, Error Handling, Technical Documentation) | **Data & Analytics** (EDA, Statistical Analysis, Data Visualisation, Vector Stores, Embeddings, Feature Pipelines) | **Developer Tools** (Git, GitHub, Docker Compose, Postman)

PROJECTS

Production-Grade Advanced RAG System (LangGraph + GCP + Groq)

- Architecting a self-correcting, cyclic RAG pipeline using LangGraph with history-aware planning and multi-step reasoning, enabling enterprise-grade hallucination reduction beyond standard retrieval approaches.
- Integrating Groq for high-speed accelerated inference to demonstrate sophisticated AI orchestration without latency trade-offs, a key production requirement for real-time applications.
- Implementing semantic re-ranking for data denoising to distinguish signal from noise in retrieved context, improving answer relevance and factual grounding.
- Deploying on Google Cloud Platform (GCP) applying cloud security, scaling, and cost-management best practices for professional AI infrastructure.

SEO-Optimised Content Generation Platform (FastAPI + AWS Bedrock)

- Built a cloud-native content generation API using FastAPI and AWS Bedrock, leveraging Claude and LLAMA LLMs to produce SEO-ready marketing copy (keywords, titles, meta descriptions, CTAs) from user-supplied business context.
- Designed structured JSON output schemas with Pydantic models, enabling reliable downstream consumption and extensibility for new content types.
- Demonstrated cloud-based LLM integration best practices: API security, error handling, response optimisation, and modular architecture.

Automated Audio Mastering Service (FastAPI + AWS)

- Implemented JWT-based authentication and CORS policies for a secure FastAPI application handling audio file upload, processing, and delivery via AWS S3.
- Integrated structured logging and error handling for S3 credential issues and file I/O failures, ensuring production reliability.

Natural Language to SQL - LLM Prototype

- Prototyped an NL-to-SQL engine using Google Gemini API and SQLite3; engineered custom prompts to handle multi-table schema awareness and edge-case query types.
- Deployed via Streamlit for rapid user testing; documented comparative analysis of AWS Bedrock and Azure OpenAI for production-grade cloud implementation.

References available upon request.